Business Intelligence

Lecture Note 4:

Data Mining

Dr Tamal Ghosh
Associate Professor
Department of CSE
Adamas University

Learning Objectives

- Define data mining as an enabling technology for business intelligence
- Understand the objectives and benefits of business analytics and data mining
- Recognize the wide range of applications of data mining
- Learn the standardized data mining processes
  - CRISP-DM
  - SEMMA
  - KDD

Learning Objectives

- Understand the steps involved in data preprocessing for data mining
- Learn different methods and algorithms of data mining
- Build awareness of the existing data mining software tools
  - Commercial versus free/open source
- Understand the pitfalls and myths of data mining

Opening Vignette…

"Data Mining Goes to Hollywood!"

- Decision situation
- Problem
- Proposed solution
- Results
- Answer & discuss the case questions

Opening Vignette:
Data Mining Goes to Hollywood!

Class No.	1	2	3	4	5	6	7	8	9
Range (in $Millions)	< 1 (Flop)	> 1 < 10	> 10 < 20	> 20 < 40	> 40 < 65	> 65 < 100	> 100 < 150	> 150 < 200	> 200 (Blockbuster)

Dependent Variable

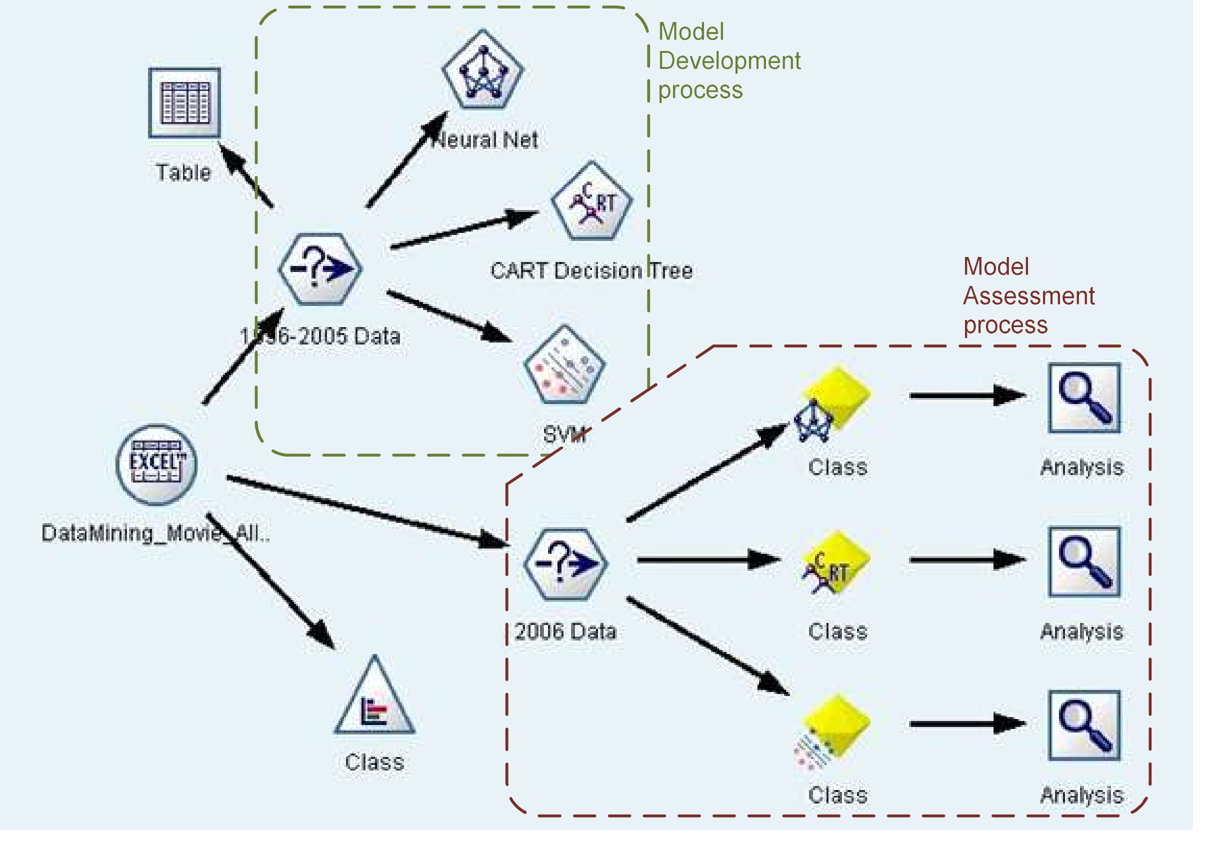
Independent Variables

Independent Variable	Number of Values	Possible Values
MPAA Rating	5	G, PG, PG-13, R, NR
Competition	3	High, Medium, Low
Star value	3	High, Medium, Low
Genre	10	Sci-Fi, Historic Epic Drama, Modern Drama, Politically Related, Thriller, Horror, Comedy, Cartoon, Action, Documentary
Special effects	3	High, Medium, Low
Sequel	1	Yes, No
Number of screens	1	Positive integer

A Typical Classification Problem

Opening Vignette:
Data Mining Goes to Hollywood!

The DM Process Map in IBM SPSS Modeler



Opening Vignette: Data Mining Goes to Hollywood!

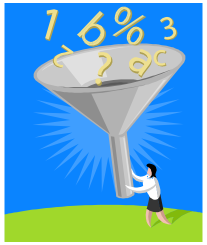
	Prediction Models					
	Individual Models			Ensemble Models		
Performance Measure	SVM	ANN	C&RT	Random Forest	Boosted Tree	Fusion (Average)
Count (*Bingo*)	192	182	140	189	187	**194**
Count (*1-Away*)	104	120	126	121	104	**120**
Accuracy (*% Bingo*)	55.49%	52.60%	40.46%	54.62%	54.05%	**56.07%**
Accuracy (*% 1-Away*)	85.55%	87.28%	76.88%	89.60%	84.10%	**90.75%**
Standard deviation	0.93	0.87	1.05	0.76	0.84	**0.63**

** Training set: 1998 – 2005 movies; Test set: 2006 movies*

Data Mining Concepts and Definitions
Why Data Mining?

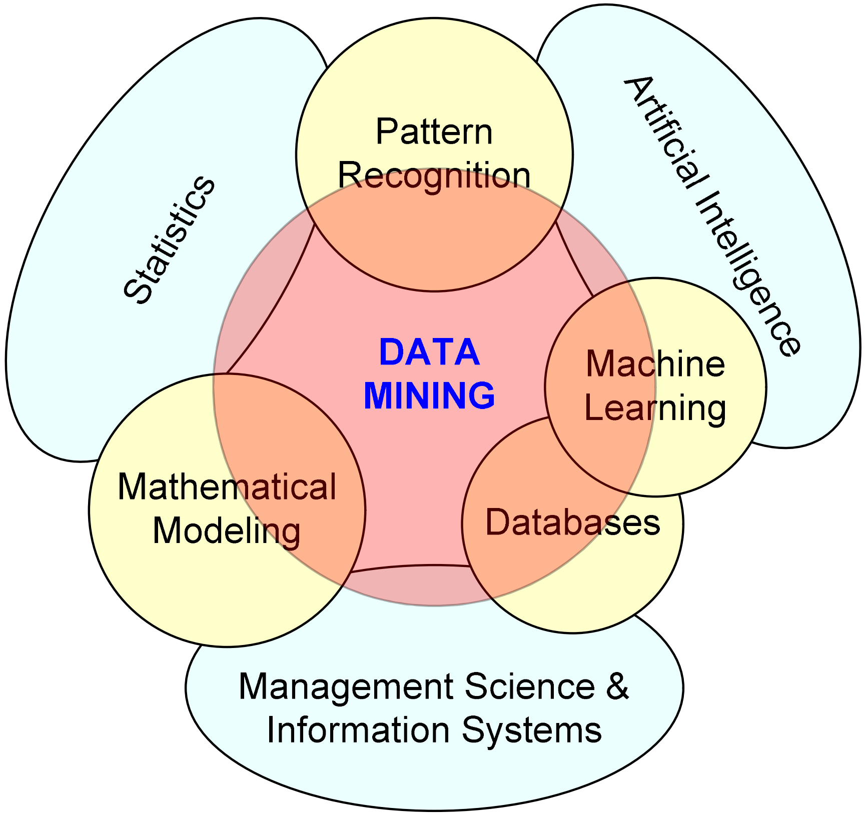
- More intense competition at the global scale
- Recognition of the value in data sources
- Availability of quality data on customers, vendors, transactions, Web, etc.
- Consolidation and integration of data repositories into data warehouses
- The exponential increase in data processing and storage capabilities; and decrease in cost
- Movement toward conversion of information resources into nonphysical form

Definition of Data Mining



- The nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data stored in structured databases
  - Fayyad et al., (1996)
- Keywords in this definition: Process, nontrivial, valid, novel, potentially useful, understandable
- Data mining: a misnomer?
- Other names: knowledge extraction, pattern analysis, knowledge discovery, information harvesting, pattern searching, data dredging

Data Mining at the Intersection of Many Disciplines

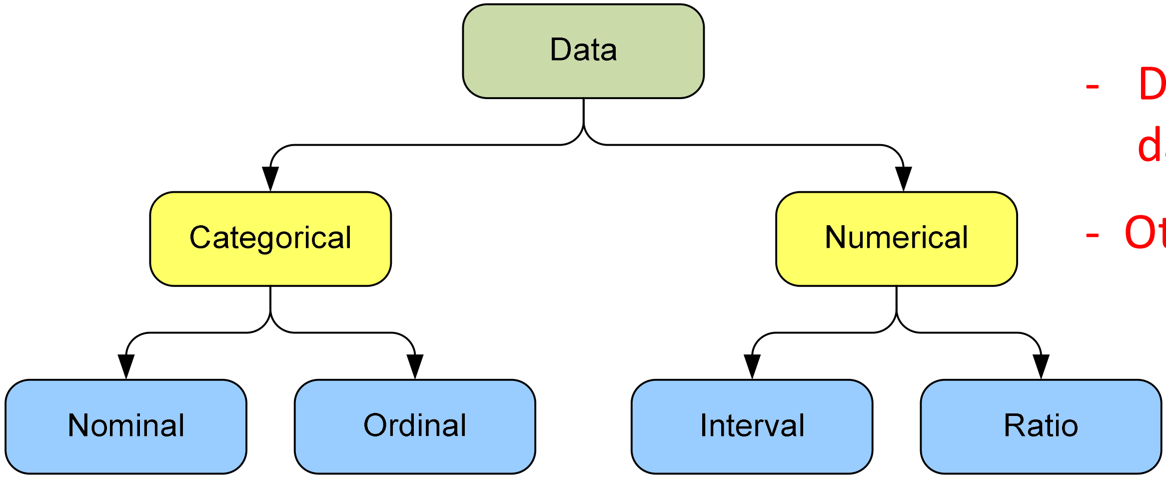


Data Mining Characteristics/Objectives

- Source of data for DM is often a consolidated data warehouse (not always!).
- DM environment is usually a client-server or a Web-based information systems architecture.
- Data is the most critical ingredient for DM which may include soft/unstructured data.
- The miner is often an end user.
- Striking it rich requires creative thinking.
- Data mining tools' capabilities and ease of use are essential (Web, Parallel processing, etc.).

Data in Data Mining

- Data: a collection of facts usually obtained as the result of experiences, observations, or experiments
- Data may consist of numbers, words, and images
- Data: lowest level of abstraction (from which information and knowledge are derived)



- DM with different data types?
- Other data types?

What Does DM Do?
How Does it Work?

- DM extracts patterns from data
  - Pattern?
    A mathematical (numeric and/or symbolic) relationship among data items
- Types of patterns
  - Association
  - Prediction
  - Cluster (segmentation)
  - Sequential (or time series) relationships

A Taxonomy for Data Mining Tasks

Data Mining	Learning Method	Popular Algorithms
Prediction	Supervised	Classification and Regression Trees, ANN, SVM, Genetic Algorithms
Classification	Supervised	Decision trees, ANN/MLP, SVM, Rough sets, Genetic Algorithms
Regression	Supervised	Linear/Nonlinear Regression, Regression trees, ANN/MLP, SVM
Association	Unsupervised	Apriory, OneR, ZeroR, Eclat
Link analysis	Unsupervised	Expectation Maximization, Apriory Algorithm, Graph-based Matching
Sequence analysis	Unsupervised	Apriory Algorithm, FP-Growth technique
Clustering	Unsupervised	K-means, ANN/SOM
Outlier analysis	Unsupervised	K-means, Expectation Maximization (EM)

Other Data Mining Tasks

- These are in addition to the primary DM tasks (prediction, association, clustering)
  - Time-series forecasting
    - Part of sequence or link analysis?
  - Visualization
    - Another data mining task?

- Types of DM
    - Hypothesis-driven data mining
    - Discovery-driven data mining

Data Mining Applications

- Customer Relationship Management
  - Maximize return on marketing campaigns
  - Improve customer retention (churn analysis)
  - Maximize customer value (cross- or up-selling)
  - Identify and treat most valued customers
- Banking & Other Financial
  - Automate the loan application process
  - Detecting fraudulent transactions
  - Maximize customer value (cross- and up-selling)
  - Optimizing cash reserves with forecasting

Data Mining Applications (cont.)

- Retailing and Logistics
  - Optimize inventory levels at different locations
  - Improve the store layout and sales promotions
  - Optimize logistics by predicting seasonal effects
  - Minimize losses due to limited shelf life
- Manufacturing and Maintenance
  - Predict/prevent machinery failures
  - Identify anomalies in production systems to optimize manufacturing capacity
  - Discover novel patterns to improve product quality

Data Mining Applications (cont.)

- Brokerage and Securities Trading
  - Predict changes on certain bond prices
  - Forecast the direction of stock fluctuations
  - Assess the effect of events on market movements
  - Identify and prevent fraudulent activities in trading
- Insurance
  - Forecast claim costs for better business planning
  - Determine optimal rate plans
  - Optimize marketing to specific customers
  - Identify and prevent fraudulent claim activities

Data Mining Applications (cont.)

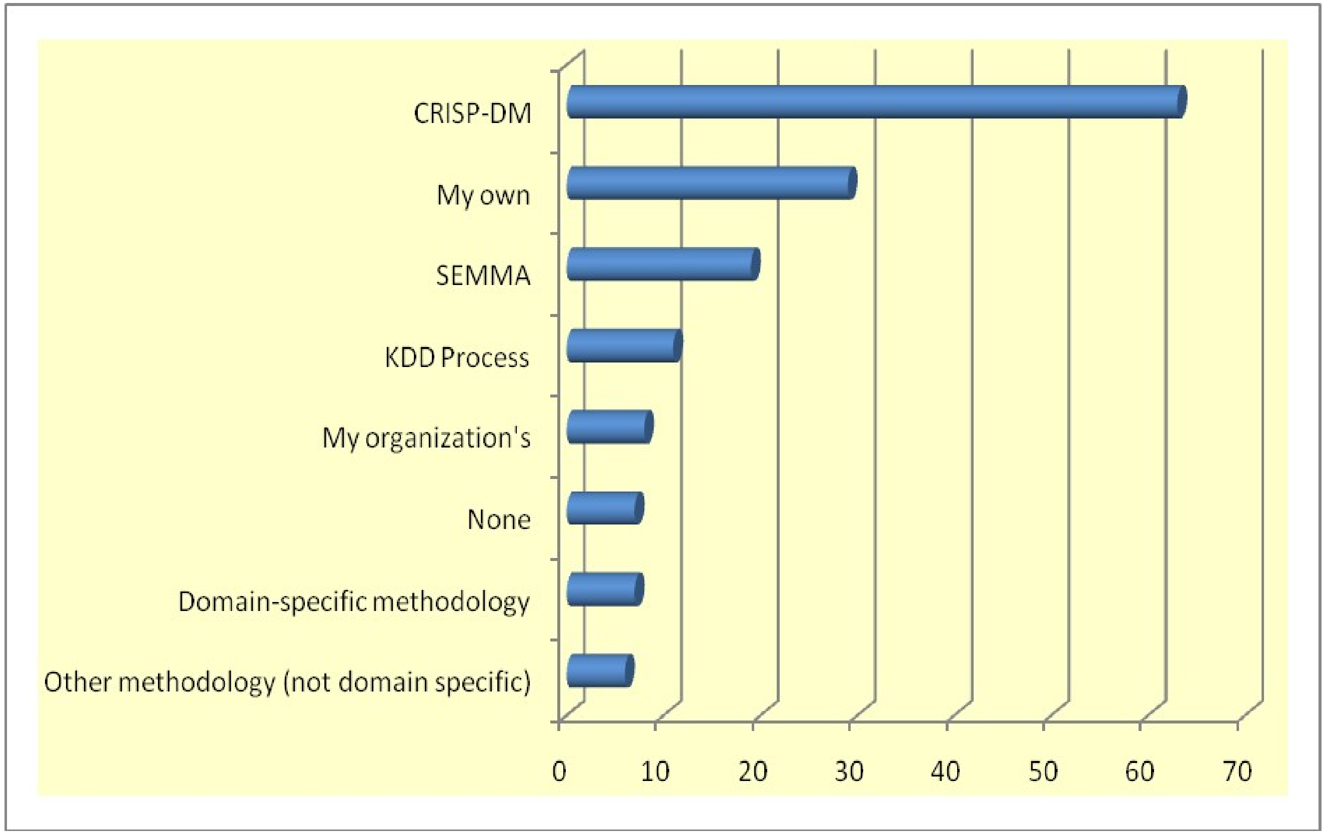
- Computer hardware and software
- Science and engineering
- Government and defense
- Homeland security and law enforcement
- Travel industry
- Healthcare
- Medicine
- Entertainment industry
- Sports
- Etc.

Highly popular application areas for data mining

Data Mining Process

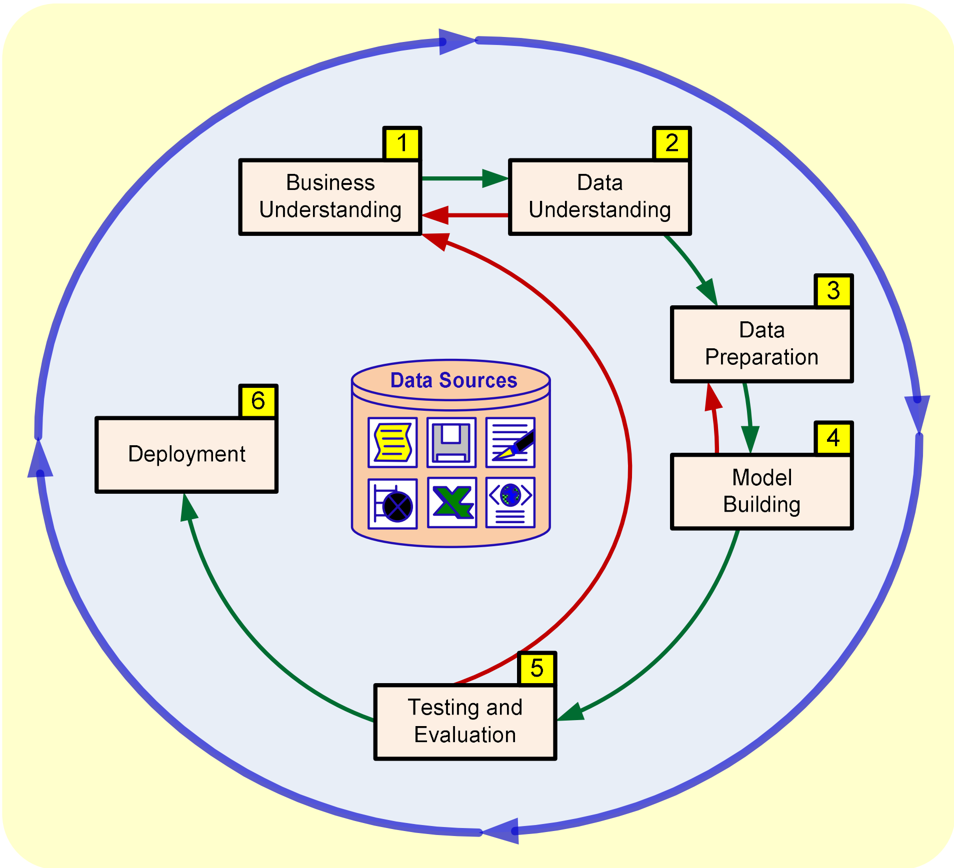
- A manifestation of best practices
- A systematic way to conduct DM projects
- Different groups have different versions
- Most common standard processes:
  - CRISP-DM (Cross-Industry Standard Process for Data Mining)
  - SEMMA (Sample, Explore, Modify, Model, and Assess)
  - KDD (Knowledge Discovery in Databases)

Data Mining Process



Source: KDNuggets.com, August 2007

Data Mining Process: CRISP-DM



Data Mining Process: CRISP-DM

Step 1: Business Understanding

Step 2: Data Understanding

Step 3: Data Preparation (!)

Step 4: Model Building

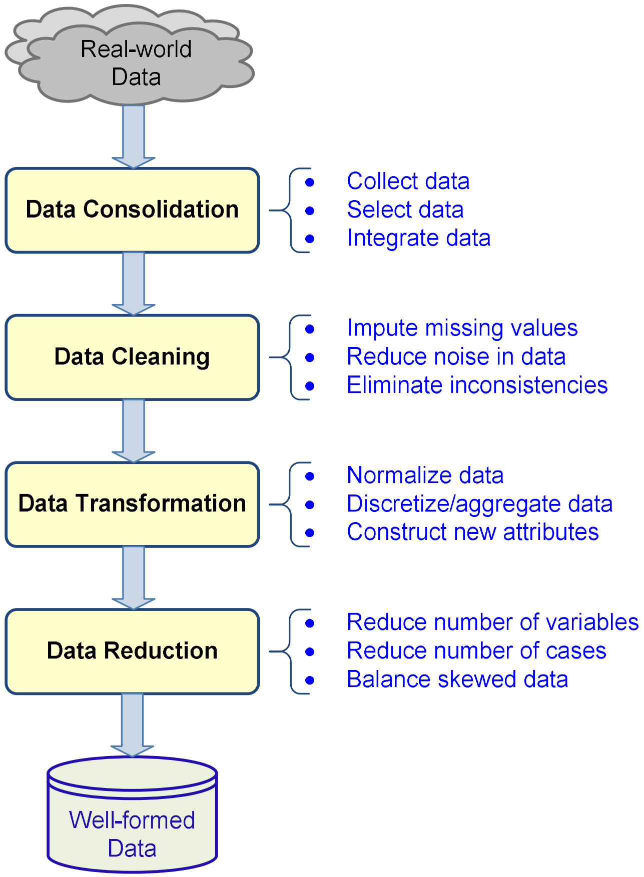
Step 5: Testing and Evaluation

Step 6: Deployment

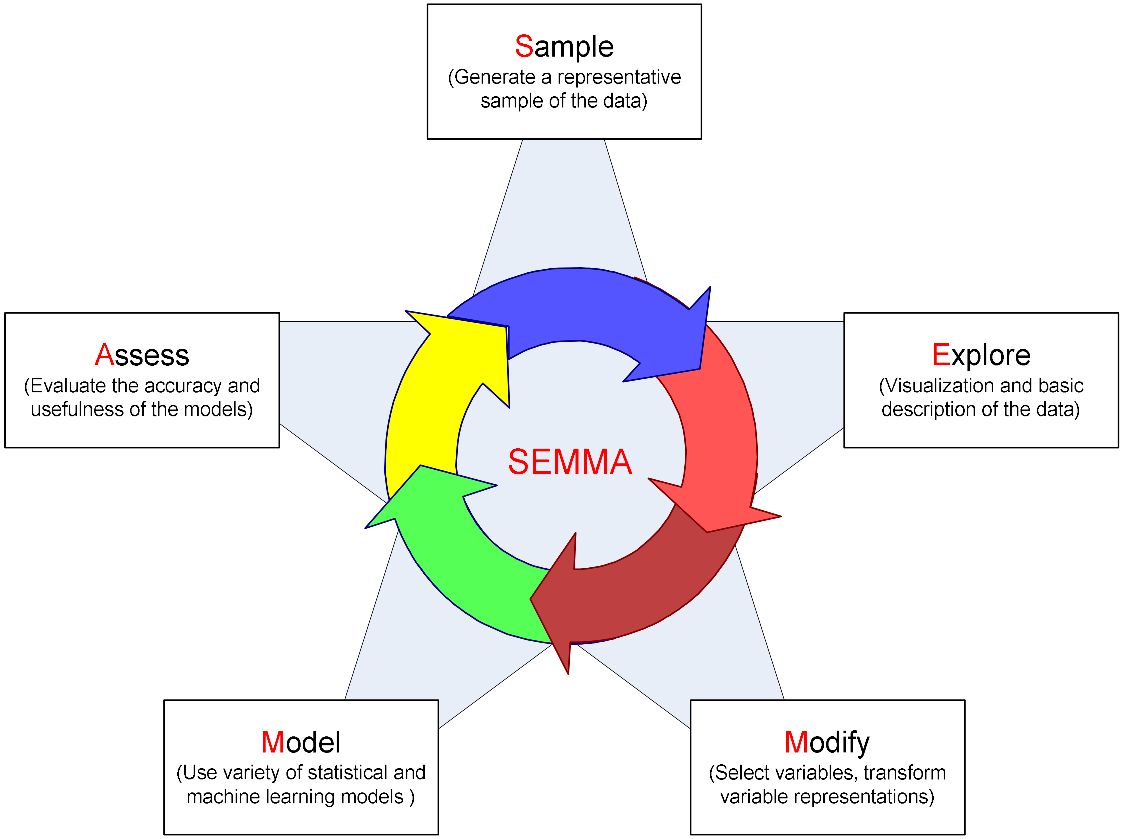
Accounts for ~85% of total project time

- The process is highly repetitive and experimental (DM: art versus science?)

Data Preparation – A Critical DM Task



Data Mining Process: SEMMA



Data Mining Methods: Classification

- Most frequently used DM method
- Part of the machine-learning family
- Employ supervised learning
- Learn from past data, classify new data
- The output variable is categorical (nominal or ordinal) in nature
- Classification versus regression?
- Classification versus clustering?

Assessment Methods for Classification

- Predictive accuracy
  - Hit rate
- Speed
  - Model building; predicting
- Robustness
- Scalability
- Interpretability
  - Transparency; ease of understanding

Accuracy of Classification Models

- In classification problems, the primary source for accuracy estimation is the confusion matrix

		True Class	
		Positive	Negative
Predicted Class	Positive	True Positive Count (TP)	False Positive Count (FP)
	Negative	False Negative Count (FN)	True Negative Count (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

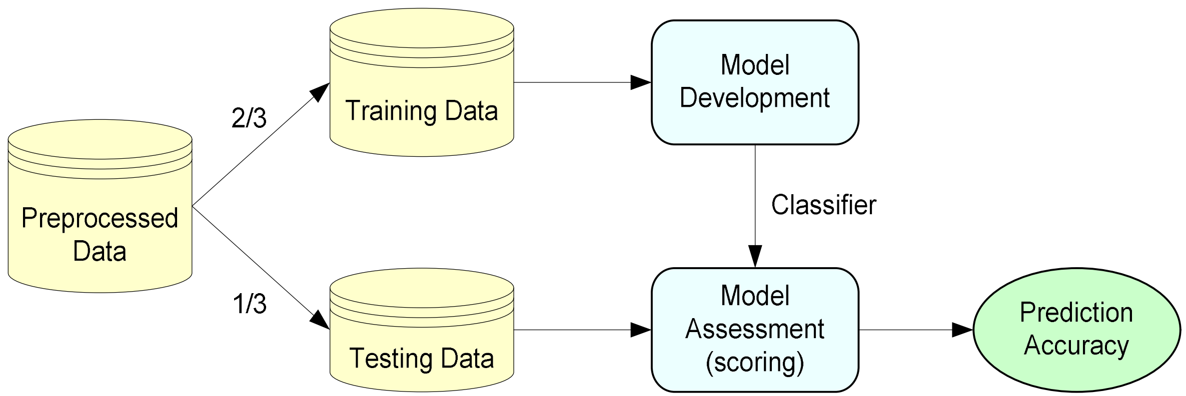
$$True\ Positive\ Rate = \frac{TP}{TP + FN}$$

$$True\ Negative\ Rate = \frac{TN}{TN + FP}$$

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

Estimation Methodologies for Classification

- Simple split (or holdout or test sample estimation)
  - Split the data into 2 mutually exclusive sets training (~70%) and testing (30%)

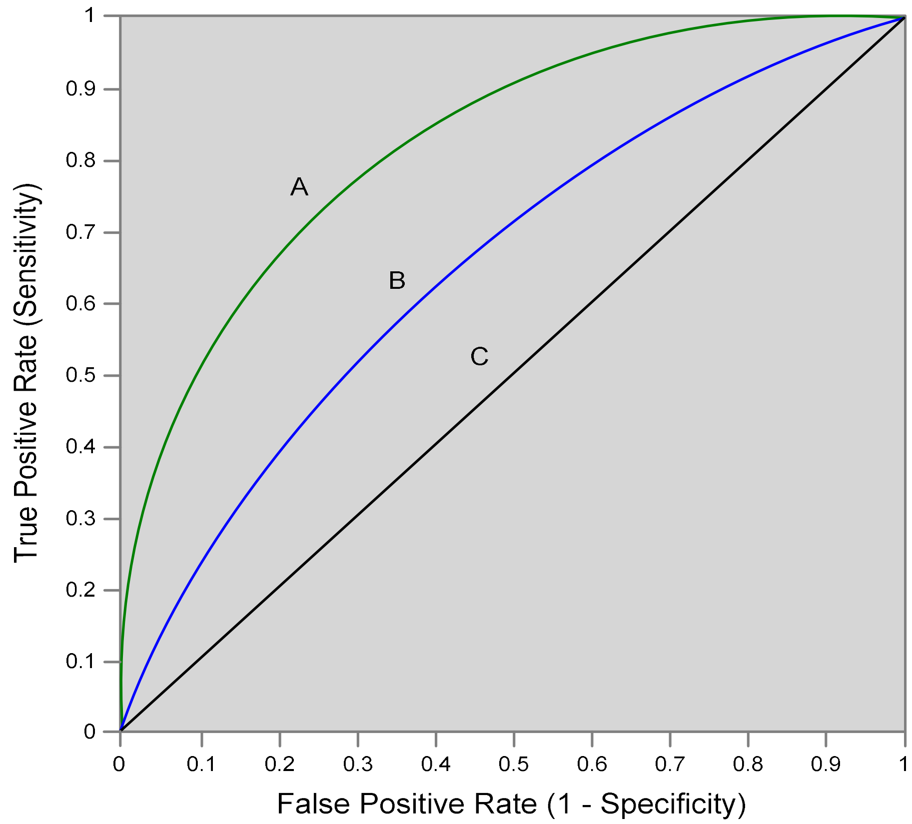


  - For ANN, the data is split into three sub-sets (training [~60%], validation [~20%], testing [~20%])

Estimation Methodologies for Classification

- *k*-Fold Cross Validation (rotation estimation)
  - Split the data into *k* mutually exclusive subsets
  - Use each subset as testing while using the rest of the subsets as training
  - Repeat the experimentation for *k* times
  - Aggregate the test results for true estimation of prediction accuracy training
- Other estimation methodologies
  - Leave-one-out, bootstrapping, jackknifing
  - Area under the ROC curve

Estimation Methodologies for Classification – ROC Curve



Classification Techniques

- Decision tree analysis
- Statistical analysis
- Neural networks
- Support vector machines
- Case-based reasoning
- Bayesian classifiers
- Genetic algorithms
- Rough sets

Decision Trees

- Employs the divide and conquer method
- Recursively divides a training set until each division consists of examples from one class

A general algorithm for decision tree building

1. ot node and assign all of the training data to it.
2. best splitting attribute.
3. ch to the root node for each value of the split. Split the data into xclusive subsets along the lines of the specific split.
4. steps 2 and 3 for each and every leaf node until the stopping eached.

Decision Trees

- DT algorithms mainly differ on
  - Splitting criteria
    - Which variable to split first?
    - What values to use to split?
    - How many splits to form for each node?
  - Stopping criteria
    - When to stop building the tree
  - Pruning (generalization method)
    - Pre-pruning versus post-pruning
- Most popular DT algorithms include
  - ID3, C4.5, C5; CART; CHAID; M5

Decision Trees

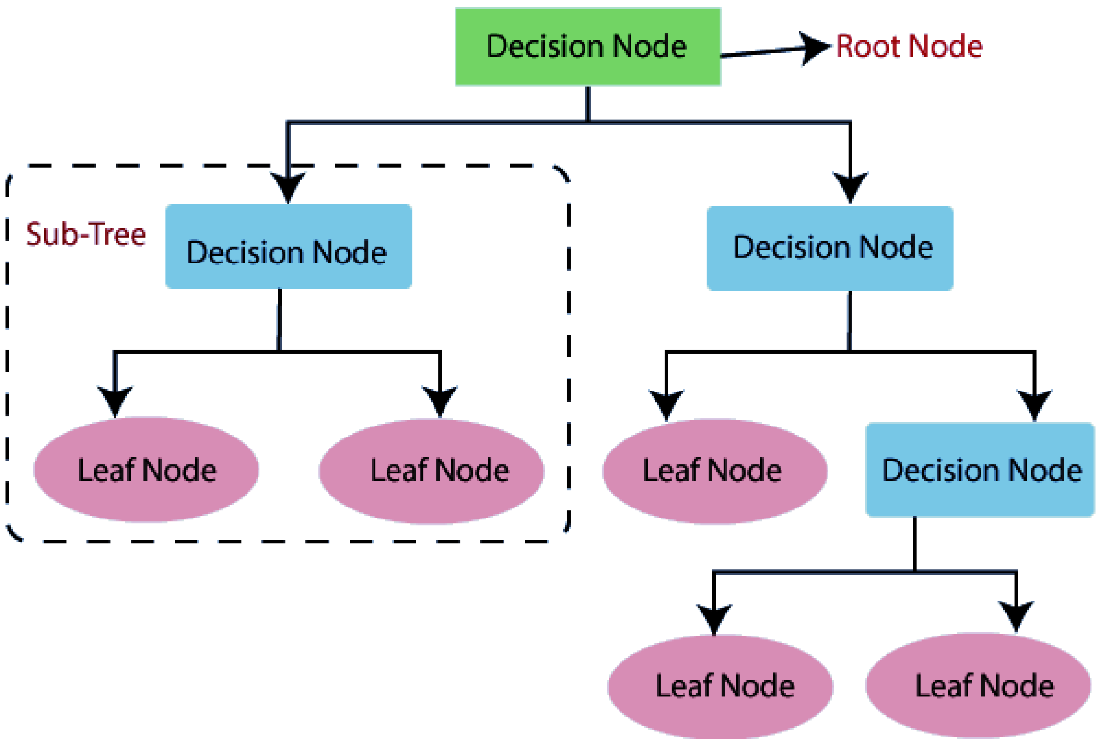
- Alternative splitting criteria
  - Gini index determines the purity of a specific class as a result of a decision to branch along a particular attribute/value
    - Used in CART
  - Information gain uses entropy to measure the extent of uncertainty or randomness of a particular attribute/value split
    - Used in ID3, C4.5, C5
  - Chi-square statistics (used in CHAID)

Decision Tree: Classification Algorithm

- *Decision Tree is a **Supervised learning technique** that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems.*
- *It is a tree-structured classifier, where **internal nodes represent the features of a dataset, branches represent the decision rules** and **each leaf node represents the outcome.***
- *The decision tree has two node types, which are the **Decision Node** and **the Leaf Node.** Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.*
- *The decisions or the test are performed based on features of the given dataset.*
- ***It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.***

Decision Tree: Classification Algorithm

- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- To build a tree, we use the **CART algorithm,** which stands for **Classification and Regression Tree algorithm.**
- A decision tree simply asks a question, and based on the answer (Yes/No), it further splits the tree into subtrees.
- The diagram at right explains the general structure of a decision tree:



Why to use Decision Trees?

- There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model. Below are the two reasons for using the Decision tree:

- **Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.**

- **The logic behind the decision tree can be easily understood because it shows a tree-like structure.**

Decision Tree Terminologies

- **Root Node:** The root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- Branch/Sub Tree: A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

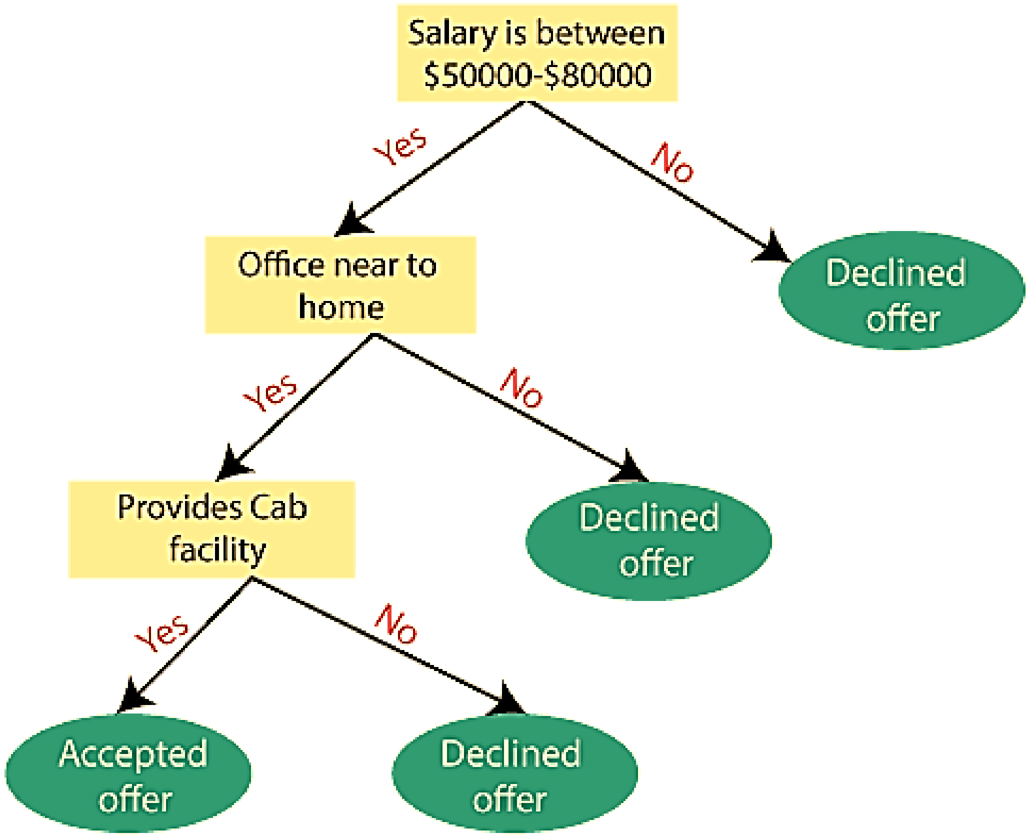
- In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of the root attribute with the record (real dataset) attribute and, based on the comparison, follows the branch and jumps to the next node.
- For the next node, the algorithm again compares the attribute value with the other sub-nodes and moves further. It continues the process until it reaches the leaf node of the tree. The complete process can be better understood using the below algorithm:

- **Step-1:** Begin the tree with the root node, says S, which contains the complete dataset.
- **Step-2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM).**
- **Step-3:** Divide the S into subsets that contain possible values for the best attributes.

Decision Tree Workflow

- **Step-4:** Generate the decision tree node, which contains the best attribute.
- **Step-5:** Recursively make new decision trees using the subsets of the dataset created in step - 3. Continue this process until a stage is reached where you cannot further classify the nodes and call the final node a leaf node.

- **Example:** Suppose there is a candidate who has a job offer and wants to decide whether he should accept the offer or Not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by



Attribute Selection Measures

- *While implementing a Decision tree, the main issue arises that how to select the best attribute for the root node and for sub-nodes. So, to solve such problems there is a technique which is called as* ***Attribute selection measure or ASM.*** *By this measurement, we can easily select the best attribute for the nodes of the tree. There are two popular techniques for ASM, which are:*

- ***Information Gain***
- ***Gini Index***

Information Gain

- *Information gain is the measurement of changes in entropy after the segmentation of a dataset based on an attribute.*
- *It calculates how much information a feature provides us about a class.*
- *According to the value of information gain, we split the node and build the decision tree.*
- *A decision tree algorithm always tries to maximize the value of information gain, and a node/attribute having the highest information gain is split first. It can be calculated using the below formula:*

Information Gain= Entropy(S)- (Weighted Avg) *Entropy(each feature)

- **Entropy: Entropy is a metric to measure the impurity in a given attribute. It specifies randomness in data. Entropy can be calculated as:**

Entropy= -P(yes)log2 P(yes)- P(no) log2 P(no)

Where,

P(yes)= probability of yes, P(no)= probability of no

Gini Index

- Gini index is a measure of impurity or purity used while creating a decision tree in the CART(Classification and Regression Tree) algorithm.
- An attribute with the low Gini index should be preferred as compared to the high Gini index.
- It only creates binary splits, and the CART algorithm uses the Gini index to create binary splits.
- Gini index can be calculated using the below formula:

$$\text{Gini Index} = 1 - \sum_j P_j^2$$

- There are many algorithms for building a decision tree. They are

1. **CART** (Classification and Regression Trees) — This makes use of Gini impurity as the metric.
2. **ID3** (Iterative Dichotomiser 3) — This uses entropy and information gain as metric.

Pruning: Getting an Optimal Decision tree

- *Pruning is a process of deleting unnecessary nodes from a tree to get the optimal decision tree.*
- A too-large tree increases the risk of overfitting, and a small tree may not capture all the important features of the dataset. Therefore, a technique that decreases the size of the learning tree without reducing accuracy is known as Pruning. There are mainly two types of tree **pruning** technology used:
- **Pre-Pruning**
- **Post-Pruning**

Pros and Cons of Using Decision Tree

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Disadvantages of the Decision Tree

- The decision tree contains lots of layers, which makes it complex.
- It may have an overfitting issue, which can be resolved using the **Random Forest algorithm.**
- For more class labels, the computational complexity of the decision tree may increase.

Example

- Consider the dataset to decide whether to play football or not:

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Example…

- Here we have four independent variables to determine the dependent variable. The independent variables are Outlook, Temperature, Humidity, and Wind. The dependent variable is whether to play football or not.
- As the first step, we have to find the parent node for our decision tree. For that follow the steps:
- ***Find the entropy of the class variable.***
- $E(S) = -[(9/14)\log(9/14) + (5/14)\log(5/14)] = 0.94$
- note: Here typically we will take log to base 2. Here total there are 14 yes/no. Out of which 9 yes and 5 no. Based on it we calculated the probability above.
- From the above data for Outlook we can arrive at the following table easily

Example.

		play		
		yes	no	total
	sunny	3	2	5
Outlook	overcast	4	0	4
	rainy	2	3	5
				14

- ***Now we have to calculate average weighted entropy***. ie, we have found the total of weights of each feature multiplied by probabilities.
- E(S, outlook) = (5/14)*E(3,2) + (4/14)*E(4,0) + (5/14)*E(2,3) = (5/14)(-(3/5)log(3/5)-(2/5)log(2/5))+ (4/14)(0) + (5/14)((2/5)log(2/5)-(3/5)log(3/5)) = 0.693
- ***The next step is to find the information gain***. It is the difference between parent entropy and average weighted entropy we found above.
- IG(S, outlook) = 0.94 - 0.693 = 0.247

Example…

- Similarly find Information gain for Temperature, Humidity, and Windy.
- IG(S, Temperature) = 0.940 - 0.911 = 0.029
- IG(S, Humidity) = 0.940 - 0.788 = 0.152
- IG(S, Windy) = 0.940 - 0.8932 = 0.048
- ***Now select the feature having the largest information gain***. Here it is Outlook. So it forms the first node(root node) of our decision t
- Now our data look as

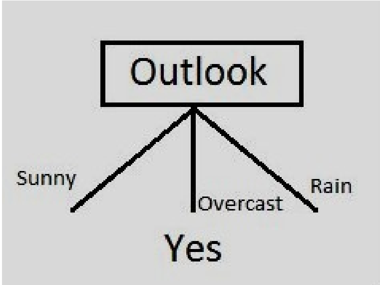
Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Example…

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Overcast	Hot	High	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- *Since overcast contains only examples of class 'Yes' we can set it as yes. That means If the outlook is overcast football will be played. Now our decision tree looks as follows.*



Example…

- The next step is to find the next node in our decision tree. Now we will find one under sunny. We have to determine which of the following Temperature, Humidi [illegible] gain.

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

- Calculate parent entropy E(sunny)
- $E(sunny) = (-(3/5)\log(3/5)-(2/5)\log(2/5)) = 0.971.$
- Now Calculate the information gain of Temperature. IG(sunny, Temperature)

		play		
		yes	no	total
	hot	0	2	2
Temperature	cool	1	1	2
	mild	1	0	1
				5

E(sunny, Temperature) = (2/5)*E(0,2) + (2/5)*E(1,1) + (1/5)*E(1,0)=2/5=0.4

Now calculate information gain.

IG(sunny, Temperature) = 0.971–0.4 =0.571

Similarly we get

IG(sunny, Humidity) = 0.971

IG(sunny, Windy) = 0.020

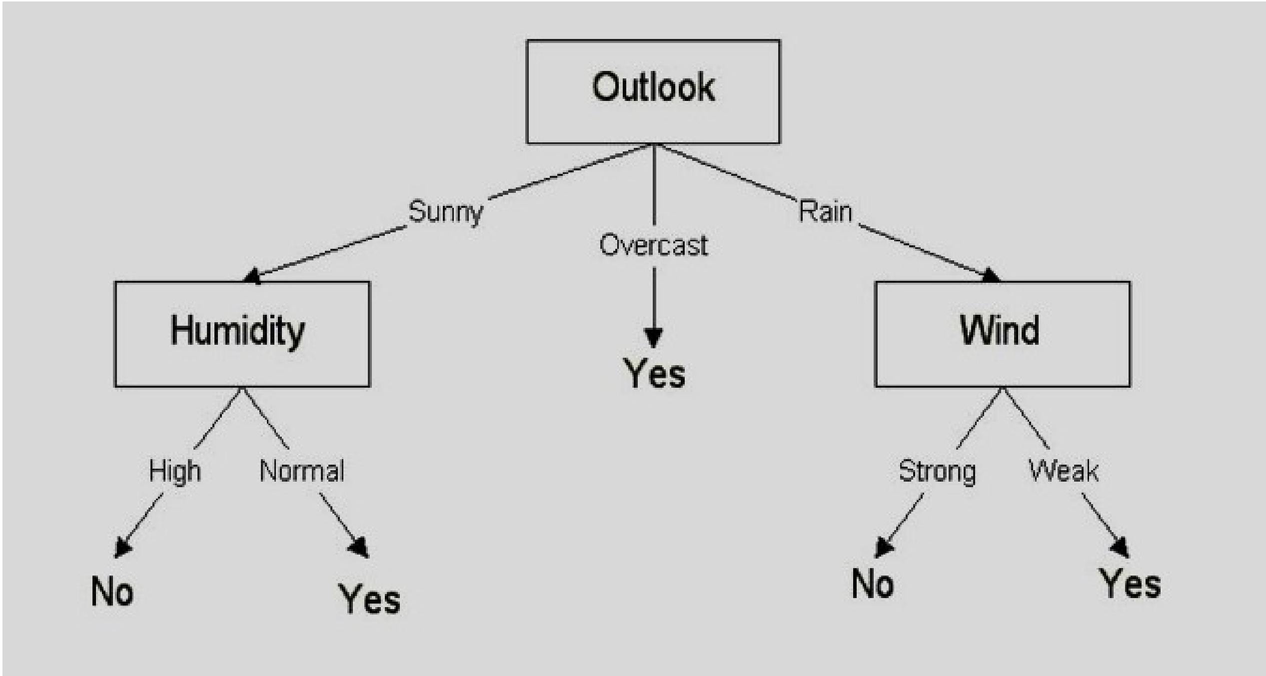
	play	
Humidity	yes	no
high	0	3
normal	2	0

Here IG(sunny, Humidity) is the largest value.

So Humidity is the node that comes under sunny.

Example

- *For humidity from the above table, we can say that play will occur if humidity is normal and will not occur if it is high. Similarly, find the nodes under rainy.*
- ***Note: A branch with entropy more than 0 needs further splitting.***
- *Finally, our decision tree will look like:*



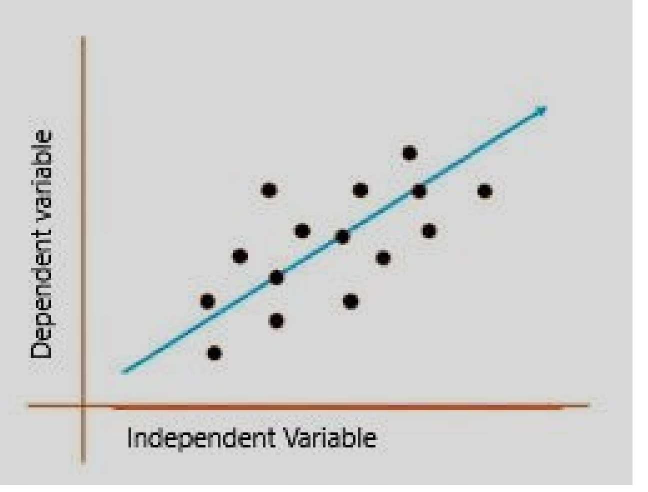
The use of the CART algorithm follows the very same procedure, except it uses Gini impurity rather than entropy.

Linear Regression

- *Linear regression predicts the relationship between two variables by assuming a linear connection between the independent and dependent variables.*
- *It seeks the optimal line that minimizes the sum of squared differences between predicted and actual values.*
- *Applied in various domains like economics and finance, this method analyzes and forecasts data trends.*
- *It can extend to multiple linear regression involving several independent variables and logistic regression, suitable for binary classification problems*
- Linear regression shows the linear relationship between the independent(predictor) variable i.e. X-axis and the dependent(output) variable i.e. Y-axis, called linear regression.

Simple Linear Regression

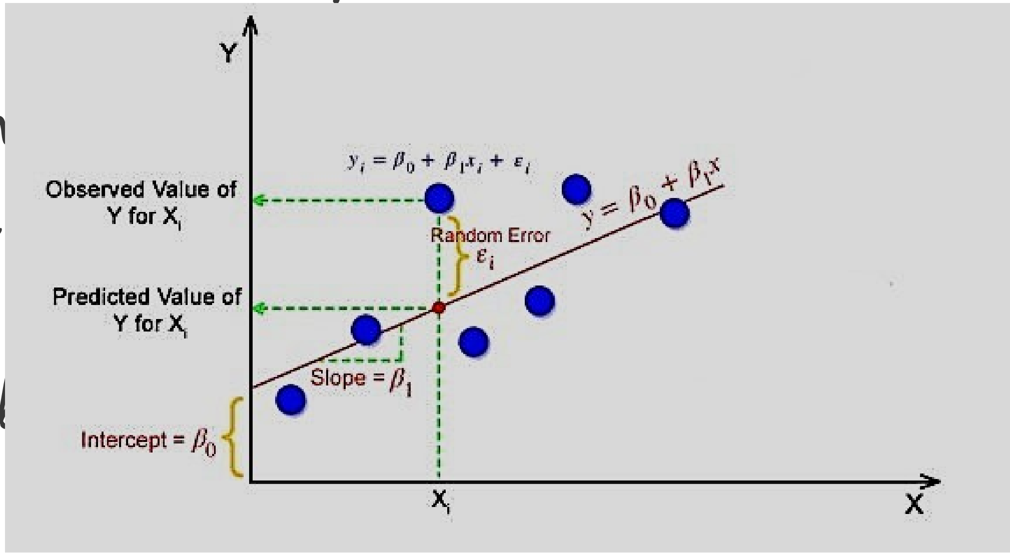
- If there is a single input variable X(independent variable), such linear regression is simple linear regression.
- In a simple linear regression, there is one independent variable and one dependent variable.
- The model estimates the slope and intercept of the line of best fit, which represents the relationship between the variables.
- The slope represents the change in the dependent variable for each unit change in the independent variable, while the intercept represents the predicted value of the dependent variable when the independent variable is zero.



Simple Regression Calculation

- To calculate best-fit line linear regression uses a traditional slope-intercept form which is given below,
- $Y = \beta_0 + \beta_1 X$
- where Y_i = Dependent variable, β_0 = constant/Intercept, β_1 = Slope/Intercept, X_i = Independent variable.

This algorithm explains the lin
relationship between the depe
(output) variable y and the
independent(predictor) varia
using a straight line $Y = \beta_0 +$



Simple Regression Calculation

- The goal of the linear regression algorithm is to get the **best values for β_0 and β_1** to find the best-fit line. The best-fit line is a line that has the least error which means the error between predicted values and actual values should be minimum.

Random Error(Residuals)

- In regression, the difference between the observed value of the dependent variable(y_i) and the predicted value(**predicted**) is called the residuals.
- $\varepsilon_i = y_{predicted} - y_i$
- **where** $y_{predicted} = \beta_0 + \beta_1 X_i$

What is the best-fit line?

- In simple terms, the best-fit line is a line that fits the given scatter plot in the best way. Mathematically, the best-fit line is obtained by minimizing the Residual Sum of Squares(RSS).

Cost Function for Linear Regression

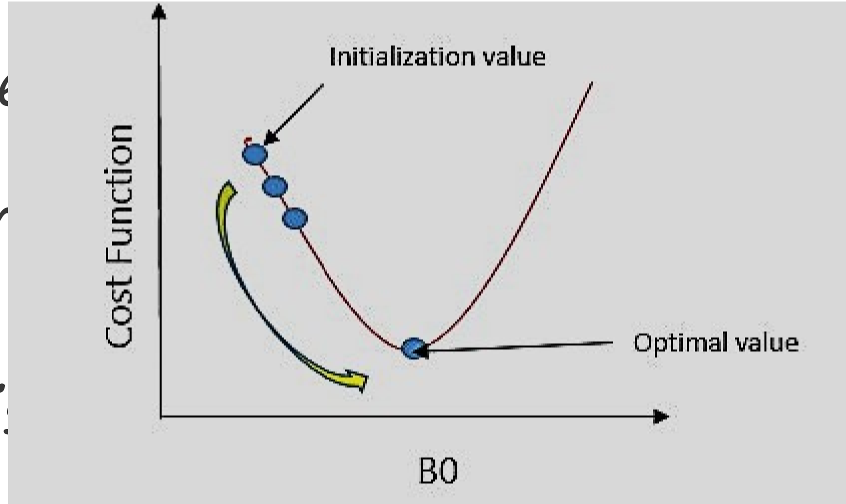
- The cost function helps to work out the optimal values for β_0 and β_1, which provides the best-fit line for the data points.
- In Linear Regression, generally **Mean Squared Error (MSE)** cost function is used, which is the average squared error that occurred between the $y_{predicted}$ and y_i.
- We cal[illegible]on $y=mx+$

$$MSE = \frac{1}{N}\sum_{i=1}^{n}(y_i - (B1x_i + B0))^2$$

- Using the MSE function, we'll update the values of β_0 and β_1 such that the MSE value settles at the minima. These parameters can be determined using the gradient descent method such that the value for the cost function is minimum.

Gradient Descent for Linear Regression

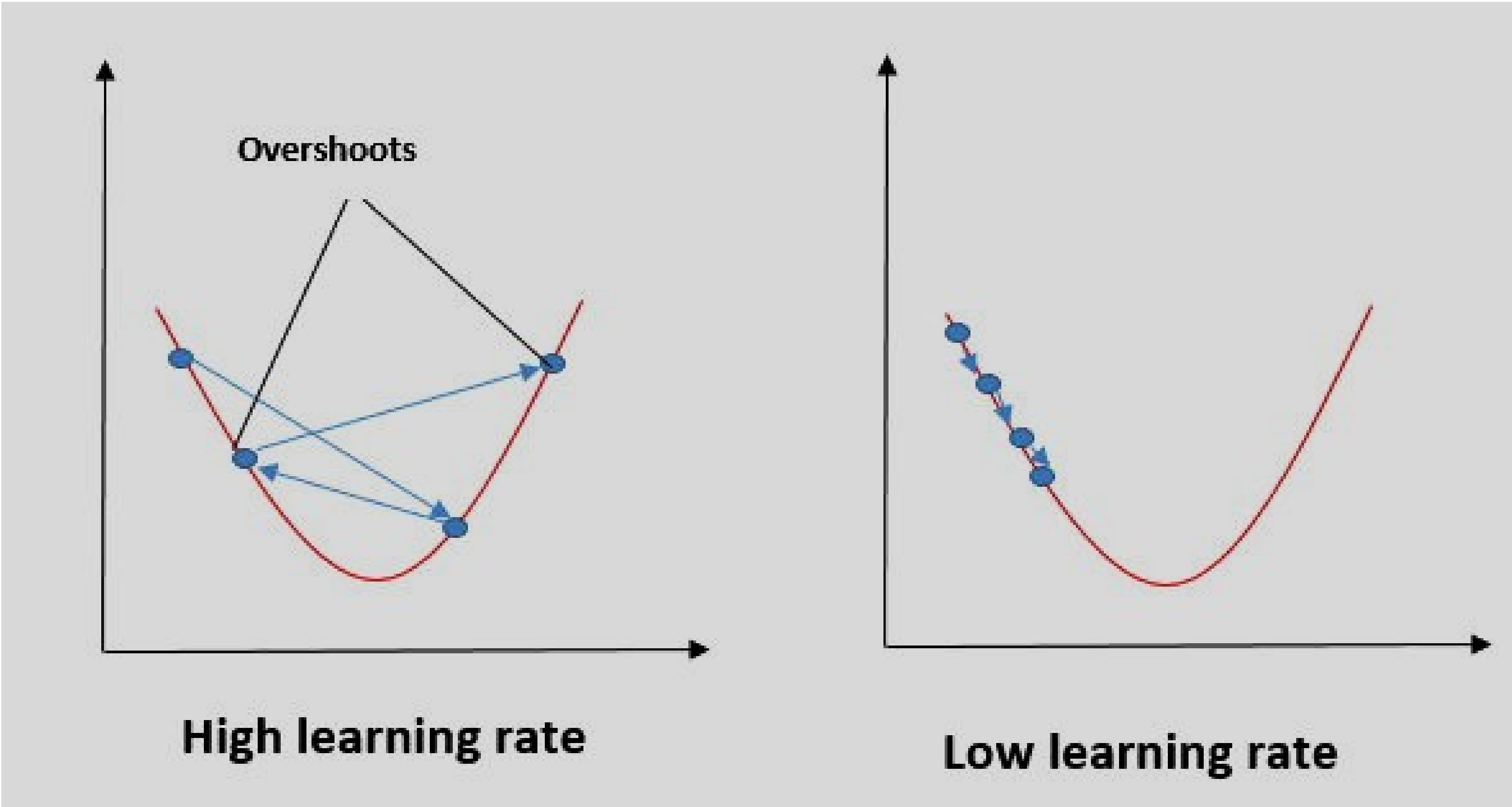
- *Gradient Descent is one of the optimization algorithms that optimize the cost function(objective function) to reach the optimal minimal solution. To find the optimum solution we need to reduce the cost function(MSE) for all data points. This is done by updating the values of B_0 and B_1 iteratively until we get an optimal solution.*
- *A regression model optimizes the gradient descent algorithm to update the coefficients of the line by reducing the cost function by randomly selecting coefficient values and then iteratively updating the values to*



Gradient Descent Example

- *Imagine a U-shaped pit and you are standing at the uppermost point in the pit, and your motive is to reach the bottom of the pit.*
- *Suppose there is a treasure at the bottom of the pit, and you can only take a discrete number of steps to reach the bottom. If you opted to take one step at a time, you would get to the bottom of the pit in the end but, this would take a longer time. If you decide to take larger steps each time, you may achieve the bottom sooner but, there's a probability that you could overshoot the bottom of the pit and not even near the bottom.*
- *In the gradient descent algorithm, the number of steps you're taking can be considered as the* ***learning rate****, and*

Gradient Descent Example



Gradient Descent Example

- To update B_0 and B_1, we take gradients from the cost function. To find these gradients, we take partial derivatives for B_0 and B_1.
- We need to minimize the cost function J. One of the ways to achieve this is to apply the batch gradient descent algorithm, the values are updated in each iteration. (The last two equations show the updating of values)
- The partial derivatives are the gradients, and they are used to update the

$$J = \frac{1}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i)^2$$

$$\frac{\partial J}{\partial B_0} = \frac{2}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i)$$

$$\frac{\partial J}{\partial B_1} = \frac{2}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i) \cdot x_i$$

$$J = \frac{1}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i)^2$$

$$\frac{\partial J}{\partial B_0} = \frac{2}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i)$$

$$\frac{\partial J}{\partial B_1} = \frac{2}{n}\sum_{i=1}^{n}(B_0 + B_1 \cdot x_i - y_i) \cdot x_i$$

$$B_0 = B_0 - \alpha \cdot \frac{2}{n}\sum_{i=1}^{n}(pred_i - y_i)$$

$$B_1 = B_1 - \alpha \cdot \frac{2}{n}\sum_{i=1}^{n}(pred_i - y_i) \cdot x_i$$

Evaluation Metrics for Linear Regression

- The strength of any linear regression model can be assessed using various evaluation metrics. These evaluation metrics usually provide a measure of how well the observed outputs are being generated by the model.
- The most used metrics are,

1. Coefficient of Determination or R-squared (R2)
2. Root Mean Squared Error (RSME) and Residual Standard Error (RSE)

Coefficient of Determination or R-squared (R2)

- R-squared is a number that explains the amount of variation that is explained/captured by the developed model. It always ranges between 0 and 1 . Overall, the higher the value of R-squared, the better the model fits the data.
- Mathematically it can be represented as,
- $$R^2 = 1 - (RSS/TSS)$$

Metrics…

- **Residual sum of Squares (RSS)** is defined as the sum of squares of the residual for each data point in the plot/data. It is the measure of the difference between the expected and the act

$$RSS = \sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2$$

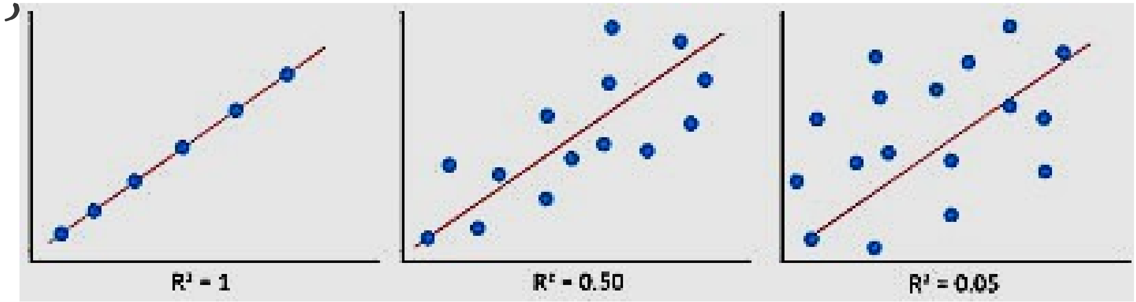
- **Total Sum of Squares (TSS)** is defined as the sum of errors of the data points from the mean of the response variable. Mathematically TSS is,

$$TSS = \sum (y_i - \bar{y}_i)^2$$

- where y hat is the mean of the sample data points.

Metrics…

- The significance of R-squared is shown by the following figures,



- The **Root Mean Squared Error** is the square root of the variance of the residuals. It specifies the absolute fit of the model to the data i.e. how close the observed data points are to the predicted value can be represented as,

$$RMSE = \sqrt{\frac{RSS}{n}} = \sqrt{\sum_{i=1}^{n} \left(y_i^{Actual} - y_i^{Predicted}\right)^2 \Big/ n}$$

Metrics…

- To make this estimate unbiased, one has to divide the sum of the squared residuals by the **degrees of freedom** rather than the total number of data points in the model. This term is then called the **Residual Standard Error(RSE)**. Mathematic

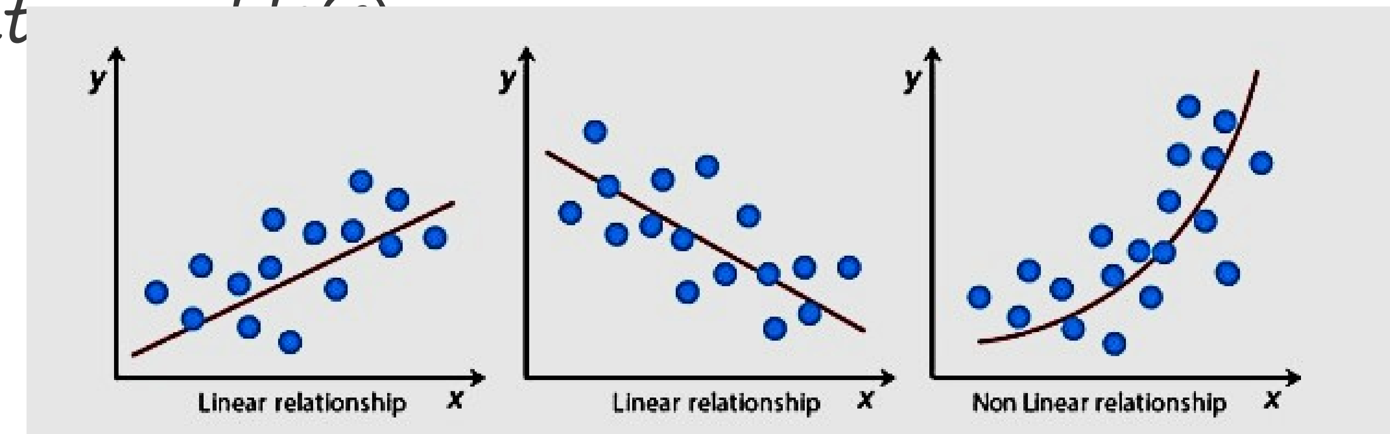
$$RSE = \sqrt{\frac{RSS}{df}} = \sqrt{\sum_{i=1}^{n}\left(y_i^{Actual} - y_i^{Predicted}\right)^2 \Big/ (n-2)}$$

- R-squared is a better measure than RSME. Because the value of Root Mean Squared Error depends on the units of the variables (i.e. it is not a normalized measure), it can change with the change in the unit of the variables.

Assumptions of Linear Regression

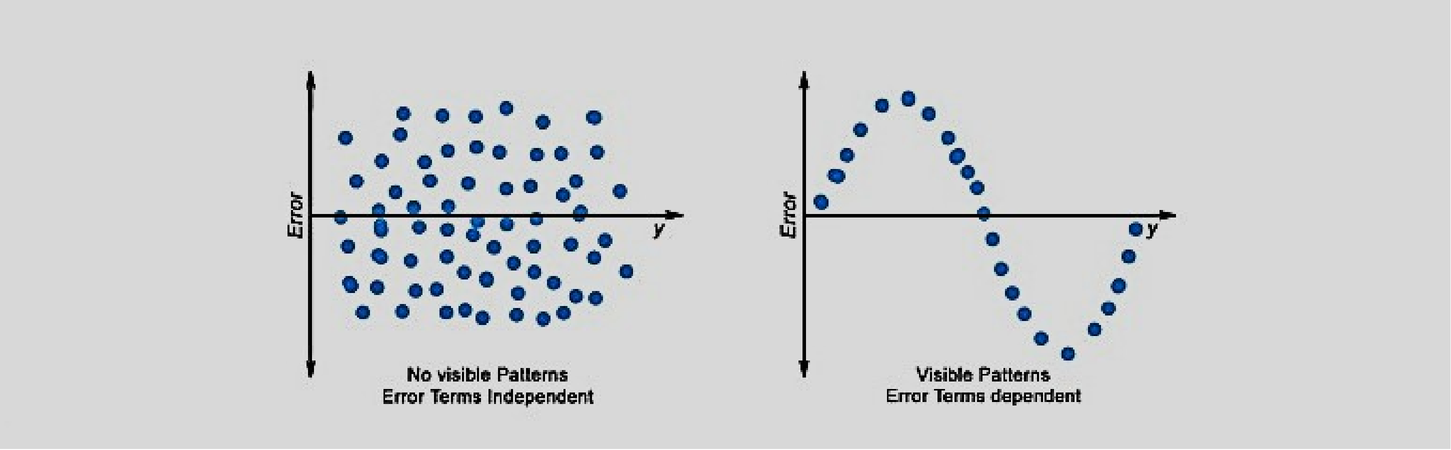
- *Regression is a parametric approach, which means that it makes assumptions about the data for the purpose of analysis. For successful regression analysis, it's essential to validate the following assumptions.*

1. **Linearity of residuals**: There needs to be a linear relationship between the dependent variable and independent variable(s).



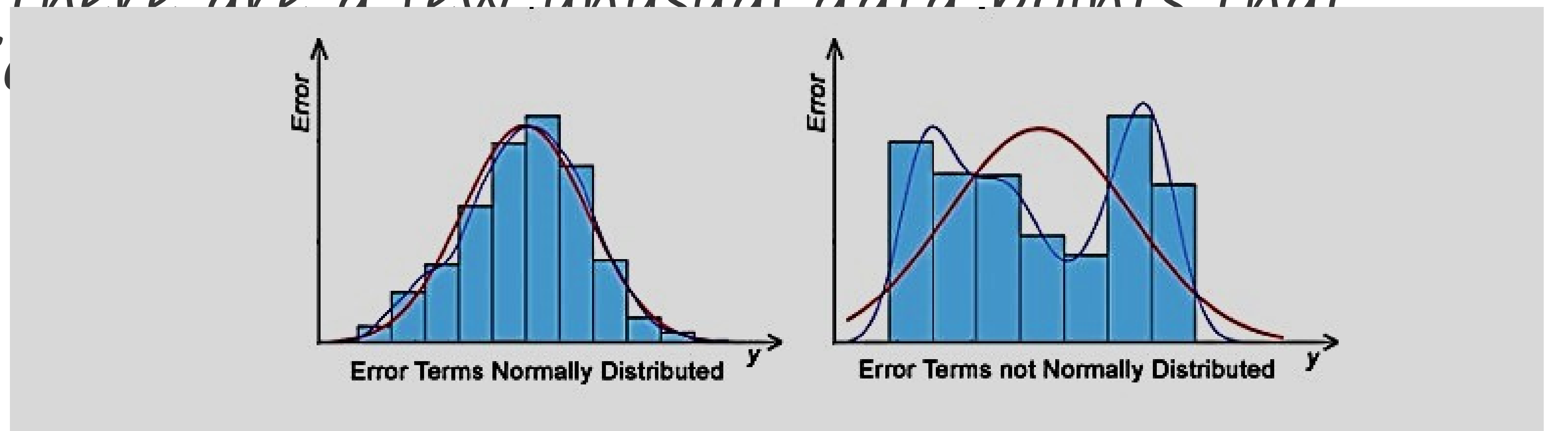
Assumptions…

- . **<u>Independence of residuals:</u>** The error terms should not be dependent on one another (like in time-series data wherein the next value is dependent on the previous one). There should be no correlation between the residual terms. The absence of this phenomenon is known as **Autocorrelation.**
- There should not terms.



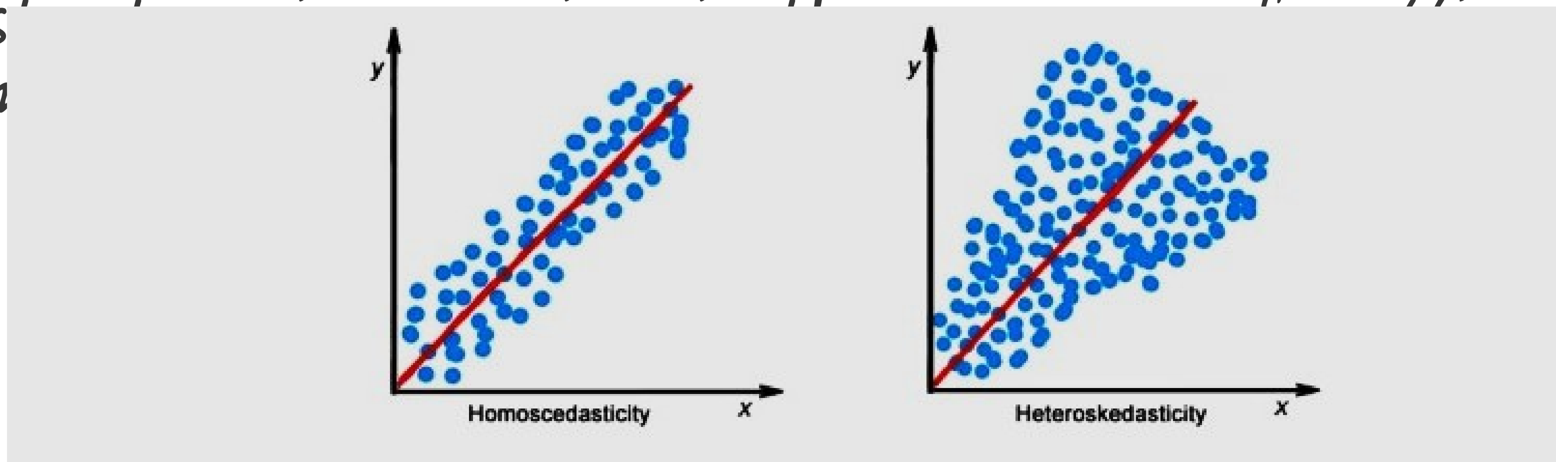
Assumptions…

- ***Normal distribution of residuals:*** The mean of residuals should follow a normal distribution with a mean equal to zero or close to zero. This is done in order to check whether the selected line is actually the line of best fit or not.
  If the error terms are non-normally distributed, suggests that there are a few unusual data points that must be studi



Assumptions…

- **The equal variance of residuals:** The error terms must have constant variance. This phenomenon is known as **Homoscedasticity**.
- The presence of non-constant variance in the error terms is referred to as **Heteroscedasticity**. Generally, non-cons
or extren



Hypothesis in Linear Regression

- Once you have fitted a straight line on the data, you need to ask, "**Is this straight line a significant fit for the data?**" Or "**Is the beta coefficient explain the variance in the data plotted?**" And here comes the idea of **hypothesis testing** on the beta coefficient. The Null and Alternate hypotheses in this case are:
- H_0: $B_1 = 0$
- H_A: $B_1 \neq 0$
- To test this hypothes test statistics for the beta coefficie

$$t = \frac{m - \mu}{s/\sqrt{n}}$$

Assessing the model fit

- Some other parameters to assess a model are:

1. **t statistic**: It is used to determine the p-value and hence, helps in determining whether the coefficient is significant or not
2. **F statistic**: It is used to assess whether the overall model fit is significant or not. Generally, the higher the value of the F-statistic, the more significant a model turns out to be.

Multiple Linear Regression

- Multiple linear regression is a technique to understand the relationship between a *single* dependent variable and *multiple* independent variables.
- The formulation for multiple linear regression is also similar to simple linear regression with
- the small change that instead of having one beta variable, you will now have betas for all the variables used. The formula is given as:
- $Y = B_0 + B_1X_1 + B_2X_2 + \ldots + B_pX_p + \varepsilon$

Considerations of Multiple Linear Regression

- *All the four assumptions made for Simple Linear Regression still hold true for Multiple Linear Regression along with a few new additional assumptions.*

1. ***Overfitting***: *When more and more variables are added to a model, the model may become far too complex and usually ends up memorizing all the data points in the training set. This phenomenon is known as the overfitting of a model. This usually leads to high training accuracy and very low test accuracy.*
2. ***Multicollinearity***: *It is the phenomenon where a model with several independent variables, may have some variables interrelated.*
3. ***Feature Selection:*** *With more variables present, selecting the optimal set of predictors from the pool of given features (many of which might be redundant) becomes an important task for building a relevant and better model.*

Multicollinearity

- *As multicollinearity makes it difficult to find out which variable is contributing towards the prediction of the response variable, it leads one to conclude incorrectly, the effects of a variable on the target variable. Though it does not affect the precision of the predictions, it is essential to properly detect and deal with the multicollinearity present in the model, as random removal of any of these correlated variables from the model causes the coefficient values to swing wildly and even change signs.*
- *Multicollinearity can be detected using the following methods.*

1. ***Pairwise Correlations:*** *Checking the pairwise correlations between different pairs of independent variables can throw useful insights into detecting multicollinearity.*

Multicollinearity…

- ***Variance Inflation Factor (VIF):*** Pairwise correlations may not always be useful as it is possible that just one variable might not be able to completely explain some other variable but some of the variables combined could be ready to do this. Thus, to check these sorts of relations between variables, one can use VIF. VIF explains the relationship of one inde[illegible]able with all the other independent variables. VIF is g[illegible]

$$\text{VIF} = \frac{1}{1-R^2}$$

- *where i refers to the i^{th} variable which is represented as a linear combination of the rest of the independent variables.*
- The common heuristic followed for the VIF values is if VIF > 10 then the value is high and it should be dropped. And if the VIF=5 then it may be valid but should be inspected first. If VIF < 5, then it is a good VIF value.

Overfitting and Underfitting in Linear Regression

- There have always been situations where a model performs well on training data but not on the test data. While training models on a dataset, overfitting, and underfitting are the most common problems faced by people.
- Before understanding overfitting and underfitting one must know about bias and variance.

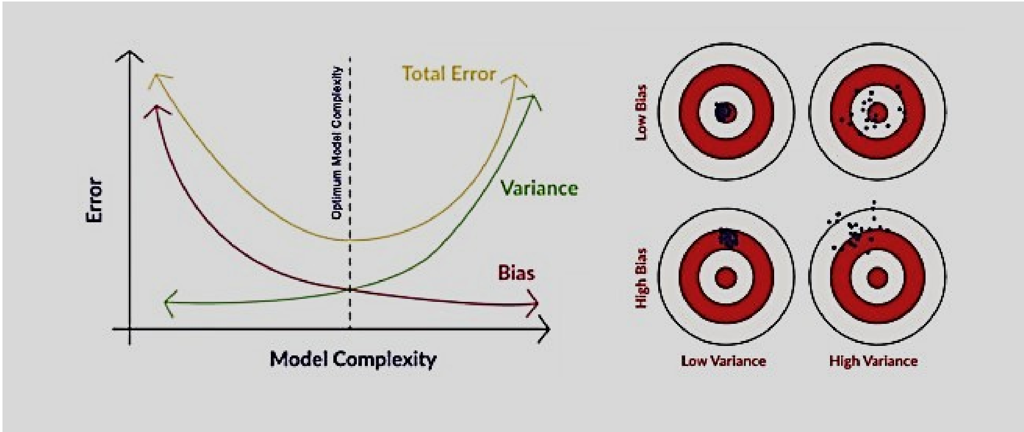
Bias:

- Bias is a measure to determine how accurate is the model likely to be on future unseen data. Complex models, assuming there is enough training data available, can do predictions accurately. Whereas the models that are too naive, are very likely to perform badly with respect to predictions. Simply, Bias is errors made by training data.

Overfitting…

- Generally, linear algorithms have a high bias which makes them fast to learn and easier to understand but in general, are less flexible. Implying lower predictive performance on complex problems that fail to meet the expected outcomes.
- **<u>Variance:</u>**
- Variance is the sensitivity of the model towards training data, that is it quantifies how much the model will react when input data is changed.
- Ideally, the model shouldn't change too much from one training dataset to the next training data, which means that the algorithm is good at picking out the hidden underlying patterns between the inputs and the output variables.
- Ideally, a model should have lower variance which means that the model doesn't change drastically after changing the training data(it is generalizable). Having a higher variance will make a model change drastically even on a small change in the training dataset.

Bias Variance Tradeoff



- In the pursuit of optimal performance, a supervised machine learning algorithm seeks to strike a balance between low bias and low variance for increased robustness.
- In the realm of machine learning, there exists an inherent relationship between bias and variance, characterized by an inverse correlation.
    - Increased bias leads to reduced variance.
    - Conversely, heightened variance results in diminished bias.

Bias…

- *Finding an equilibrium between bias and variance is crucial, and algorithms must navigate this trade-off for optimal outcomes.*
- *In practice, calculating precise bias and variance error terms is challenging due to the unknown nature of the underlying target function.*
- *Now, let's delve into the nuances of overfitting and underfitting.*

Overfitting

- When a model learns each and every pattern and noise in the data to such extent that it affects the performance of the model on the unseen future dataset, it is referred to as ***overfitting***. The model fits the data so well that it interprets noise as patterns in the data.
- When a model has low bias and higher variance it ends up memorizing the data and causing overfitting. Overfitting causes the model to become specific rather than generic. This usually leads to high training accuracy and very low test accuracy.
- Detecting overfitting is useful, but it doesn't solve the actual problem. There are several ways to prevent overfitting, which are stated below:
  - Cross-validation
  - If the training data is too small to train add more relevant and clean data.
  - If the training data is too large, do some feature selection and remove unnecessary features.
  - Regularization

Underfitting

- Underfitting is not often discussed as often as overfitting is discussed. When the model fails to learn from the training dataset and is also not able to generalize the test dataset, is referred to as ***underfitting***. This type of problem can be very easily detected by the performance metrics.
- When a model has high bias and low variance it ends up not generalizing the data and causing underfitting. It is unable to find the hidden underlying patterns from the data. This usually leads to low training accuracy and very low test accuracy. The ways to prevent underfitting are stated below,
- Increase the model complexity
- Increase the number of features in the training data
- Remove noise from the data.

Hands on

- Let’s take the dataset as an example,

Month	Money Spent (X)	Avg(X) – X	[Avg(X) – X]^2	Sales (Y)	Avg(Y) – Y	[Avg(X)-X * Avg(Y)-Y]
1	2000	1300	1,690,000	5000	7600	11,400,000
2	3000	300	90,000	10000	2600	780,000
3	4000	-700	490,000	15000	-2400	1,680,000
4	2500	800	640,000	8000	4600	3,680,000
5	5000	-1700	2,890,000	25000	-12400	21,080,000
AVG(X)	3300			12600		
		SUM XX	5,800,000		SUM YY	37,918,000

Hands on…

- Find the average of the
- X variable, in this case, it is the Money Spent = 3300
- Y variable, in this case, it is Sale = 12600
- Measure the difference between the Average X and individual X
- Square the difference and add the result: **SUM XX = 5, 800,000**
- Multiple the between Avg(X)-X and Avg(Y)-Y and add the results: **SUM XY = 37,918,000**
- To calculate the Slope of the Line, divide the SUM XY by SUM XX
- b (Slope of the line) = 37,918,000 / 5,800,000 = 6.53758621
- To calculate the y-intercept subtract Avg(Y) from Slope * AVG(X)
- a (y-intercept) = 12600 – 6.53758621* 3300 = 8,974.0345
- So, now you have a Simple Linear Regression
- **Y = a + bX**
- **Y = 8,974.0345 + 6.53758621 * X**

Cluster Analysis for Data Mining

- Used for automatic identification of natural groupings of things
- Part of the machine-learning family
- Employ unsupervised learning
- Learns the clusters of things from past data, then assigns new instances
- There is no output variable
- Also known as segmentation

Cluster Analysis for Data Mining

- Clustering results may be used to
  - Identify natural groupings of customers
  - Identify rules for assigning new cases to classes for targeting/diagnostic purposes
  - Provide characterization, definition, labeling of populations
  - Decrease the size and complexity of problems for other data mining methods
  - Identify outliers in a specific domain (e.g., rare-event detection)

Cluster Analysis for Data Mining

- Analysis methods
  - Statistical methods (including both hierarchical and nonhierarchical), such as *k*-means, *k*-modes, and so on.
  - Neural networks (adaptive resonance theory [ART], self-organizing map [SOM])
  - Fuzzy logic (e.g., fuzzy c-means algorithm)
  - Genetic algorithms
- Divisive versus Agglomerative methods

Cluster Analysis for Data Mining

- How many clusters?
  - There is no “truly optimal” way to calculate it
  - Heuristics are often used
    - Look at the sparseness of clusters
    - Number of clusters = $(n/2)^{1/2}$ (n: no of data points)
    - Use Akaike information criterion (AIC)
    - Use Bayesian information criterion (BIC)
- Most cluster analysis methods involve the use of a distance measure to calculate the closeness between pairs of items.
  - Euclidian versus Manhattan (rectilinear) distance

Cluster Analysis for Data Mining

- *k*-Means Clustering Algorithm
  - *k* : pre-determined number of clusters
  - Algorithm (Step 0: determine value of *k*)

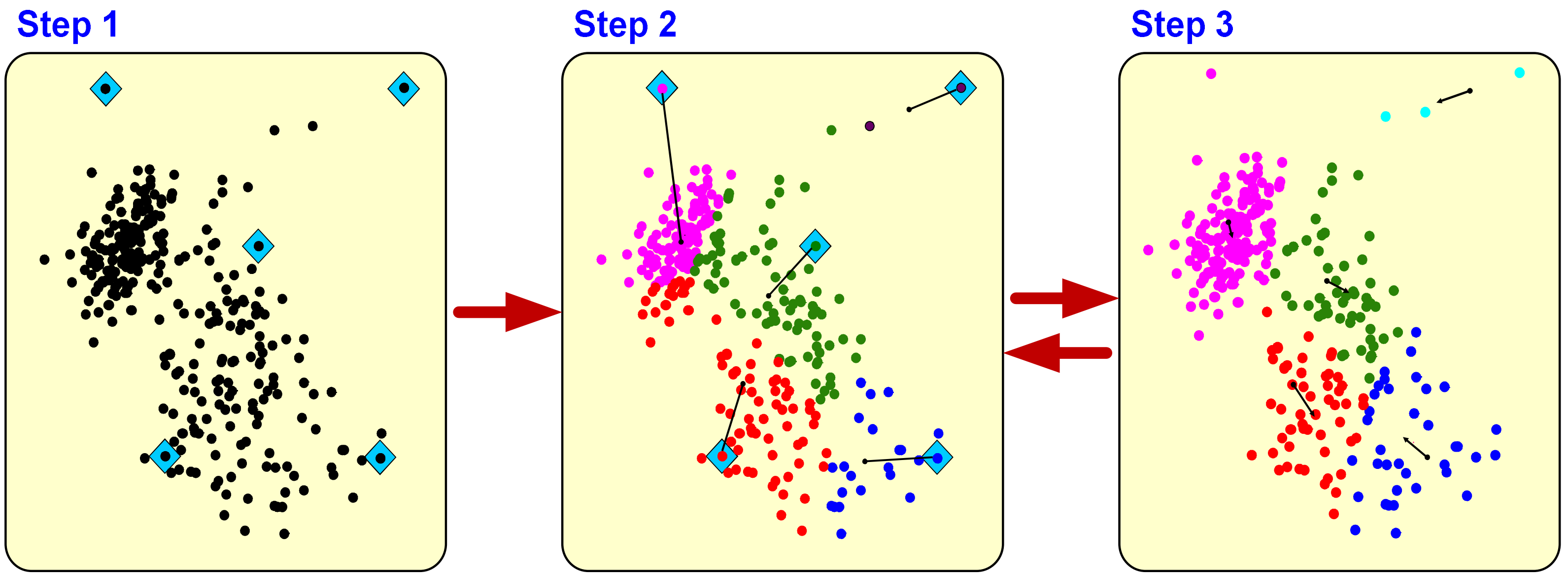
  Step 1: Randomly generate *k* random points as initial cluster centers.

  Step 2: Assign each point to the nearest cluster center.

  Step 3: Re-compute the new cluster centers.

  Repeat steps 3 and 4 until some convergence criterion is met (usually that the assignment of points to clusters becomes stable).

Cluster Analysis for Data Mining - *k*-Means Clustering Algorithm



Association Rule Mining

- A very popular DM method in business
- Finds interesting relationships (affinities) between variables (items or events)
- Part of machine learning family
- Employs unsupervised learning
- There is no output variable
- Also known as market basket analysis
- Often used as an example to describe DM to ordinary people, such as the famous “relationship between diapers and beers!”

Association Rule Mining

- Input: the simple point-of-sale transaction data
- Output: Most frequent affinities among items
- Example: according to the transaction data…

  "Customer who bought a laptop computer and a virus protection software, also bought extended service plan 70 percent of the time"
- How do you use such a pattern/knowledge?
  - Put the items next to each other for ease of finding
  - Promote the items as a package (do not put one on sale if the other(s) are on sale)
  - Place items far apart from each other so that the customer has to walk the aisles to search for it, and by doing so potentially see and buy other items

Association Rule Mining

- Representative applications of association rule mining include
  - In business: cross-marketing, cross-selling, store design, catalog design, e-commerce site design, optimization of online advertising, product pricing, and sales/promotion configuration
  - In medicine: relationships between symptoms and illnesses; diagnosis and patient characteristics and treatments (to be used in medical DSS); and genes and their functions (to be used in genomics projects)

Association Rule Mining

- Are all association rules interesting and useful?

A Generic Rule: **X $\Rightarrow$ Y [S%, C%]**

X, Y: products and/or services
X: Left-hand-side (LHS)
Y: Right-hand-side (RHS)
S: Support: how often **X** and **Y** go together
C: Confidence: how often **Y** go together with the **X**

Example: {Laptop Computer, Antivirus Software} $\Rightarrow$ {Extended Service Plan} [30%, 70%]

Association Rule Mining

- Algorithms are available for generating association rules
  - Apriori
  - Eclat
  - FP-Growth
  - + Derivatives and hybrids of the three
- The algorithms help identify the frequent item sets, which are, then converted to association rules

Association Rule Mining

- Apriori Algorithm
  - Finds subsets that are common to at least a minimum number of the itemsets
  - Uses a bottom-up approach
    - frequent subsets are extended one item at a time (the size of frequent subsets increases from one-item subsets to two-item subsets, then three-item subsets, and so on)
    - groups of candidates at each level are tested against the data for minimum support (*see the figure*) →

Association Rule Mining

- Apriori Algorithm

Raw Transaction Data

Transaction No	SKUs (Item No)
1	1, 2, 3, 4
1	2, 3, 4
1	2, 3
1	1, 2, 4
1	1, 2, 3, 4
1	2, 4

One-item Itemsets

Itemset (SKUs)	Support
1	3
2	6
3	4
4	5

Two-item Itemsets

Itemset (SKUs)	Support
1, 2	3
1, 3	2
1, 4	3
2, 3	4
2, 4	5
3, 4	3

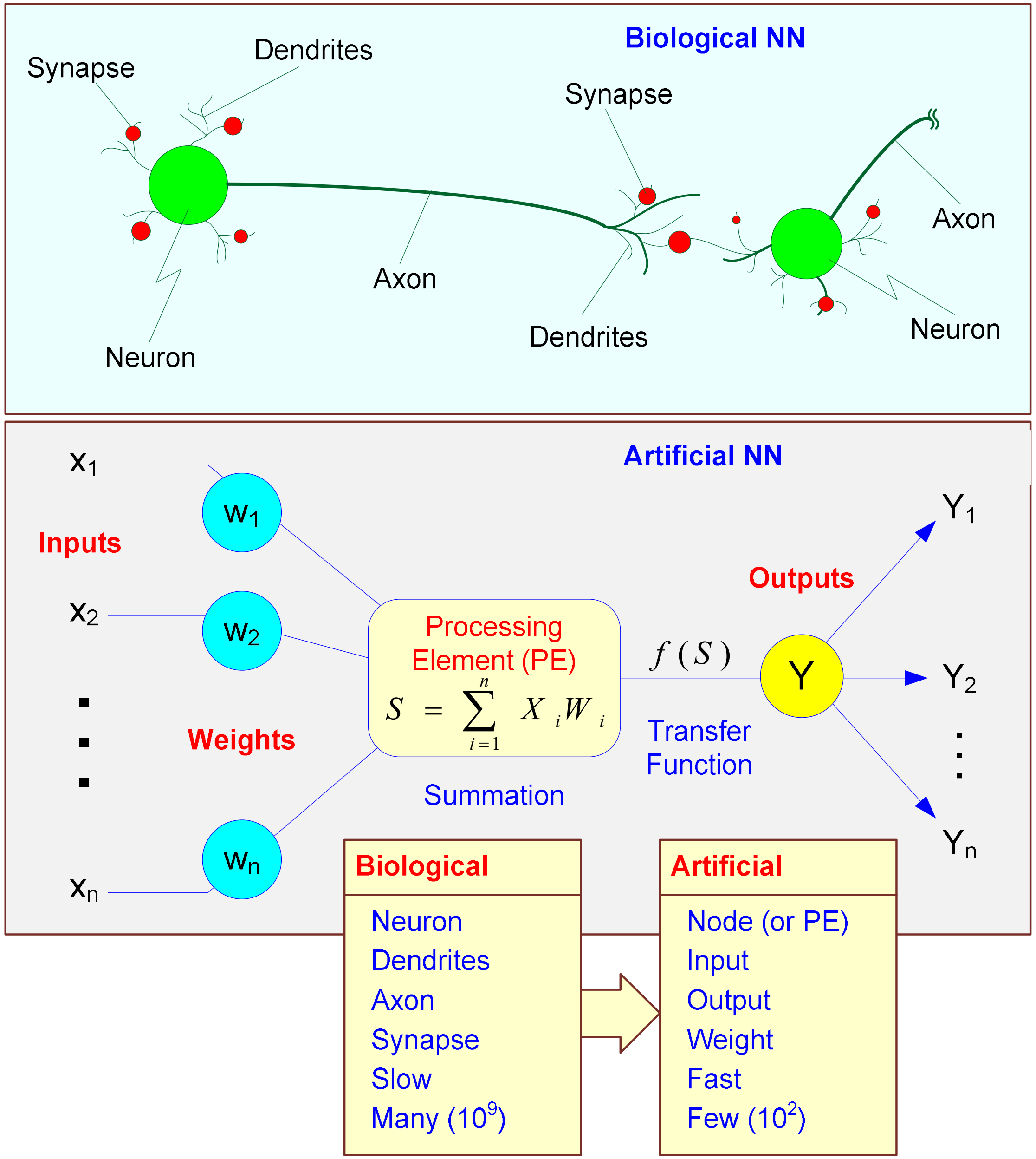
Three-item Itemsets

Itemset (SKUs)	Support
1, 2, 4	3
2, 3, 4	3

Artificial Neural Networks for Data Mining

- Artificial neural networks (ANN or NN) is a brain metaphor for information processing
- a.k.a. Neural Computing
- Very good at capturing highly complex non-linear functions!
- Many uses – prediction (regression, classification), clustering/segmentation
- Many application areas – finance, medicine, marketing, manufacturing, service operations, information systems, etc.

Biological versus Artificial Neural Networks

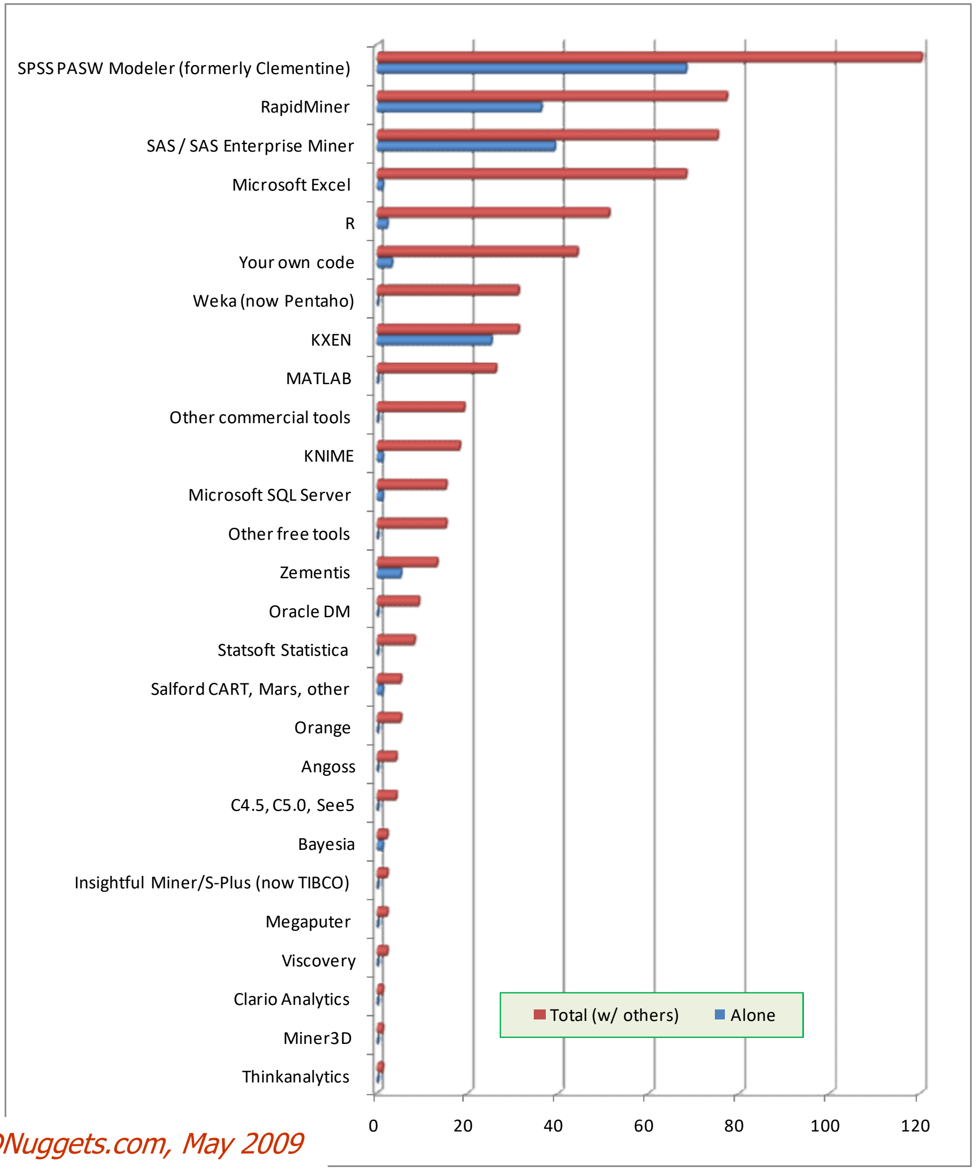


Elements/Concepts of ANN

- Processing element (PE)
- Information processing
- Network structure
  - Feedforward vs. recurrent vs. multi-layer…
- Learning parameters
  - Supervised/unsupervised, backpropagation, learning rate, momentum
- ANN Software – NN shells, integrated modules in comprehensive DM software, …

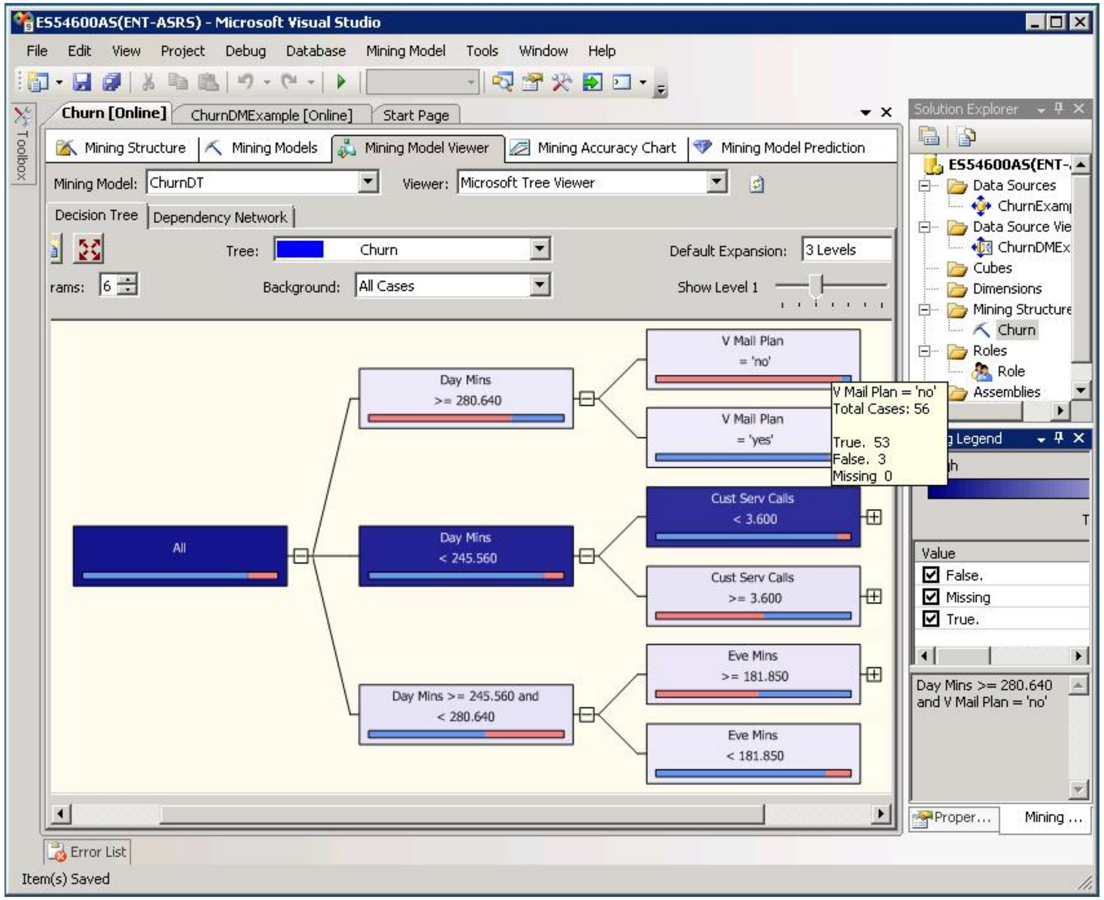
Data Mining Software

- Commercial
  - IBM SPSS Modeler (formerly Clementine)
  - SAS – Enterprise Miner
  - IBM – Intelligent Miner
  - StatSoft – Statistica Data Miner
  - … many more
- Free and/or Open Source
  - RapidMiner
  - Weka
  - … many more



Source: KDNuggets.com, May 2009

Data Mining in MS SQL Server 2008



Data Mining Myths

- Data mining …
  - provides instant solutions/predictions.
  - is not yet viable for business applications.
  - requires a separate, dedicated database.
  - can only be done by those with advanced degrees.
  - is only for large firms that have lots of customer data.
  - is another name for good-old statistics.

Common Data Mining Blunders

1. Selecting the wrong problem for data mining
2. Ignoring what your sponsor thinks data mining is and what it really can/cannot do
3. Not leaving sufficient time for data acquisition, selection and preparation
4. Looking only at aggregated results and not at individual records/predictions
5. Being sloppy about keeping track of the data mining procedure and results

Common Data Mining Mistakes

6. Ignoring suspicious (good or bad) findings and quickly moving on
7. Running mining algorithms repeatedly and blindly, without thinking about the next stage
8. Naively believing everything you are told about the data
9. Naively believing everything you are told about your own data mining analysis
10. Measuring your results differently from the way your sponsor measures them

End of the Chapter

- Questions, comments